Virtual Memory, Paging and Page Replacement

LumoraAI Presentation Demo • 02 Operating Systems

Virtual memory

• Virtual memory gives each process the abstraction of a large address space while allowing the operating system to map virtual pages to physical frames and, when necessary, secondary storage.

Paging

• Paging divides virtual memory into fixed-size pages and physical memory into frames. A page table records mappings. A translation-lookaside buffer (TLB) caches recent address translations to reduce page-table lookup overhead.

Page faults and replacement

• A page fault occurs when a referenced page is not currently available in the required physical frame. The operating system must locate the page, possibly evict another page, update mappings and restart the interrupted access.

• FIFO, LRU and Optimal are classic replacement strategies. Optimal is a theoretical benchmark because it requires knowledge of future references.

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